



Immunization Planning App User Guide

Turn district coverage data into a costed, session-level vaccination plan — import observations, find the gap, assign teams, and export a reviewable microplan.

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1 What the app does

The Immunization Planning app answers a practical question: **how many eligible children in each settlement are still missing a vaccine dose, and what teams, doses, and budget are needed to reach them?**

You import settlement-level observations (eligible children and children already vaccinated), the app joins them to your uploaded administrative boundaries, and you work through four tabs:

- **Need** — coverage, remaining gap, and unassigned children per area and settlement, with a map.
- **Sessions** — assign fixed, outreach, or mobile sessions to settlements; record actual delivery later.
- **Resources** — deterministic dose, team-day, and budget calculations from your assumptions.
- **Export** — an Excel planning workbook, a printable session plan, and a JSON backup.

Key terms

Observation	One settlement's numbers for a specific vaccine, dose, cohort, and period: eligible children and children already vaccinated.
Remaining gap	Eligible children minus vaccinated children, minus any delivery you have recorded in sessions.
Session	A planned visit: settlement, date, team, delivery mode, and the number of children assigned.
Workspace	Your currently loaded set of administrative boundaries. Plans are stored against that exact boundary set.
Scenario copy	An independent duplicate of a plan, used to compare alternative assumptions or allocations.

All calculations are deterministic

Every number in this app comes from arithmetic you can audit — there is no AI estimation in coverage, gap, dose, or budget figures. The formulas are printed in each section of this guide.

2 Before you start

You need two things:

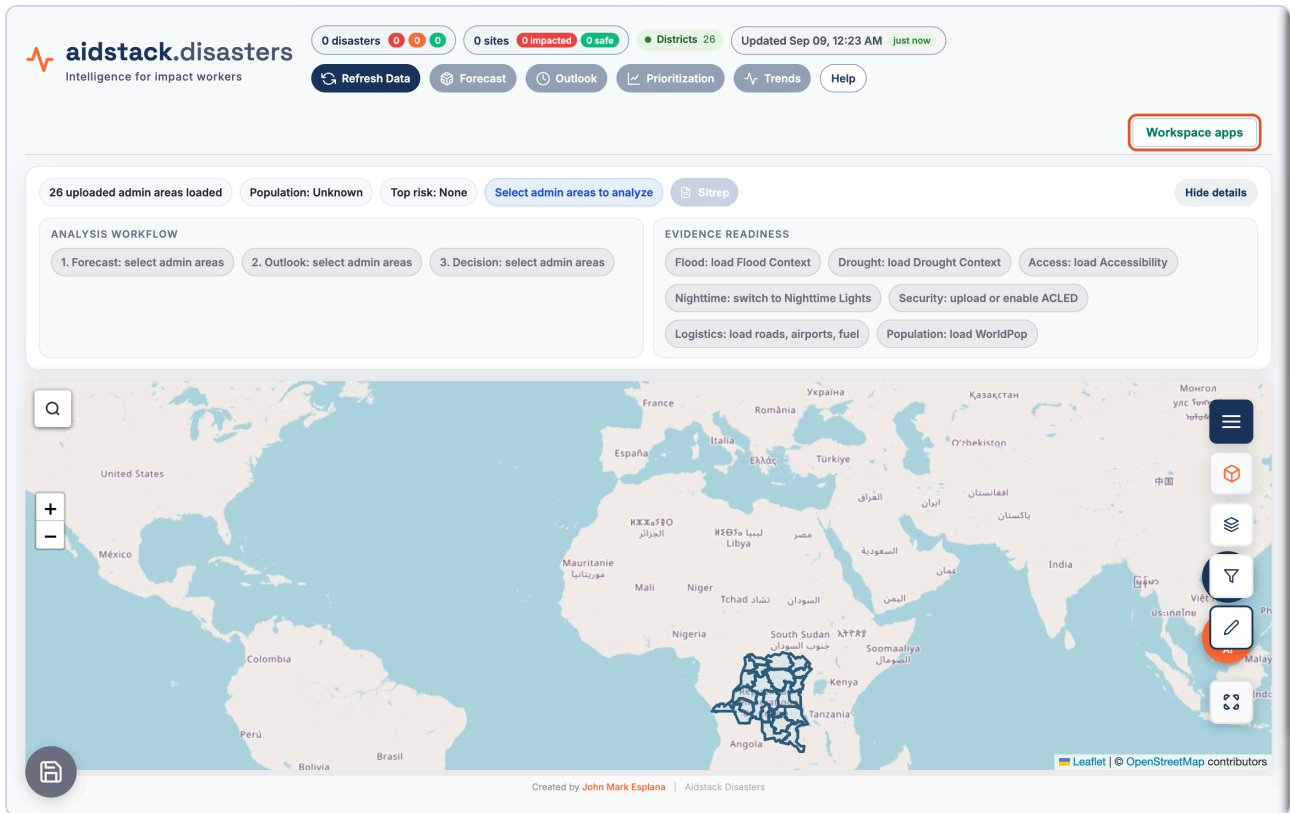
1. **Administrative boundaries loaded in the map workspace.** Upload your shapefile or GeoJSON (for example, DRC provinces or health zones) using the normal boundary upload. The app cannot be enabled without boundaries, because every observation must join to an area.
2. **A settlement dataset** as CSV or Excel — one row per settlement with eligible and vaccinated child counts. Section 5 describes the exact columns, and the app can generate a starter template.

Plans are tied to your boundary set

A saved plan belongs to the exact boundaries that were loaded when it was created. If you later replace the boundary file, existing plans will not open against the new geography. Finish or export plans before switching boundary files.

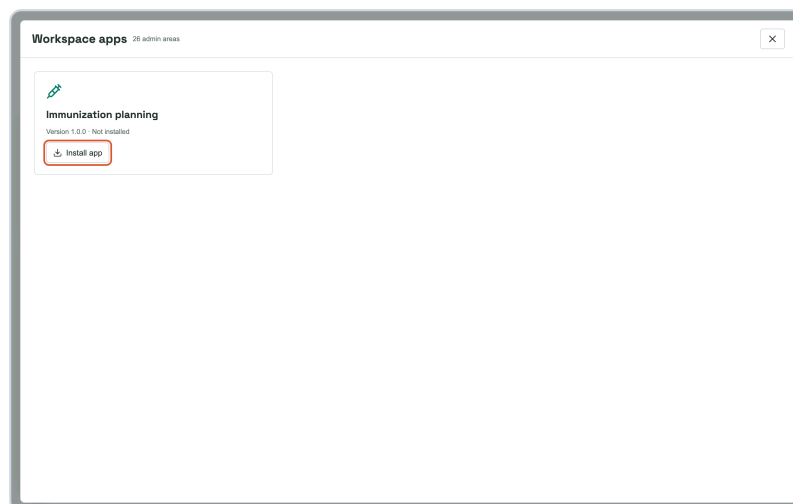
3 Open and install the app

1. In the main workspace, confirm your boundaries are loaded (the **Districts** badge in the header shows the count).
2. Click **Workspace apps** (highlighted below).



The main workspace with 26 DRC provinces loaded. The Workspace apps button opens the app catalog.

3. In the catalog, click **Install app** on the Immunization planning card, then **Open**.



Installing enables the app for this boundary workspace. Disabling it later keeps your saved plans.

4 Create a plan

Every plan starts with six required fields. They are not decoration — they define what your numbers mean, and they appear on every export so reviewers know exactly what they are looking at.

Field	What to enter	Example
Plan name	A name your team will recognize.	North Kivu measles catch-up Q3
Vaccine / antigen	The vaccine this plan covers. One plan = one antigen and dose.	Measles (MCV)
Dose	Which dose in the schedule.	1
Eligible age / cohort	The age group your counts refer to.	6–59 months
Observation period	The reporting period of the imported data.	2026-Q3
Data source / revision	Where the numbers came from, and which version.	EPI district reports, Aug 2026

← Immunization planning 26 admin areas

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Open saved plan... + New plan Restore backup

New immunization plan

Plan name

North Kivu measles catch-up Q3

Vaccine / antigen

Measles (MCV)

Dose

1

Eligible age / cohort

6-59 months

Observation period

2026-Q3

Data source / revision

EPI district reports, Aug 2026

Import CSV / Excel Use workspace sites (0) Template

The new plan form. Below the metadata are three ways to load observations: import a file, reuse workspace sites, or download the CSV template.

Keep cohorts consistent

Coverage math is only valid when every row uses the same cohort, dose, and period definitions. If some districts report 6–11 months and others 6–59 months, resolve that before importing — do not mix them in one plan.

5 Prepare and import your data

The importer accepts CSV, XLSX, and XLS files up to 25 MB and 50,000 rows. The first row must be a single header row with unique column names. Use **Template** in the app to download a correctly ordered starter file.

Column	Required	Description
settlement_id	Yes	A unique ID per settlement within its area (village code, site code...).
settlement	Yes	Settlement name, as it should appear in plans and exports.
target_population	Yes	Eligible children in the cohort. Whole number, zero allowed.
children_vaccinated	Yes	Children in the cohort already vaccinated with this dose. Whole number, zero allowed, must not exceed the target.
area_code	One of these two	The admin area code matching a field in your boundary file (for example a PCODE).
latitude + longitude	One of these two	Decimal-degree coordinates; the settlement is joined to the boundary polygon containing the point.

Do not paste data into the template around its header

The template's column order is fixed. If you paste another file's rows — including its header line — underneath the template header, every value shifts into the wrong column and validation rejects all rows. Import your data file directly, with exactly one header row.

Import and map columns

1. Click **Import CSV / Excel** and select your file. The app reads it in the background; you can cancel a long import.
2. Check the column mapping. Columns whose names match the template are mapped automatically; fix any that are not.
3. If you join by coordinates, leave **Admin area code** unmapped and map both coordinate columns. If you join by code, pick the matching **Boundary join field** (for example `PCODE`) so codes are compared against the right attribute of your boundary file.

The screenshot shows the 'Immunization planning' interface with 26 admin areas. The 'New immunization plan' section is active, showing fields for Plan name, Vaccine / antigen, Dose, Eligible age / cohort, Observation period, and Data source / revision. Below these are buttons for 'Import CSV / Excel', 'Use workspace sites (0)', and 'Template'. The '12 observations' section shows a mapping table with columns for Boundary join field, Settlement ID, Settlement name, Admin area code, Eligible children, Vaccinated children, Latitude (optional), and Longitude (optional). The mapping is as follows:

Boundary join field	Settlement ID	Settlement name	Admin area code	Eligible children	Vaccinated children	Latitude (optional)	Longitude (optional)
Platform area ID	settlement_id	settlement	Select column...	target_population	children_vaccinated	latitude	longitude

At the bottom of the mapping section is a 'Validate observations' button.

Mapping for a coordinate-based join: area code left unmapped, latitude and longitude mapped, counts auto-matched by name.

6 Validate observations

Click **Validate observations**. Every row is checked before it can enter a plan — nothing is silently corrected. Valid rows are counted; rejected rows are listed with their source row number and every problem found, and can be downloaded for cleanup.

← Immunization planning 26 admin areas

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Open saved plan... + New plan Restore backup

New immunization plan

Plan name

North Kivu measles catch-up Q3

Vaccine / antigen

Measles (MCV)

Dose

1

Eligible age / cohort

6-59 months

Observation period

2026-Q3

Data source / revision

EPI district reports, Aug 2026

Import CSV / Excel Use workspace sites (0) Template

12 observations

Boundary join field

Platform area ID

Settlement ID

settlement_id

Settlement name

settlement

Admin area code

Select column...

Eligible children

target_population

Vaccinated children

children_vaccinated

Latitude (optional)

latitude

Longitude (optional)

longitude

Validate observations

12 valid rows; 0 excluded rows.

☒ The accepted rows represent non-overlapping eligible cohorts for this vaccine, dose, and observation period. Excluded rows will not contribute to totals.

Create plan from 12 valid rows

A clean validation: 12 valid rows, 0 excluded. The confirmation checkbox is your sign-off that the rows describe non-overlapping eligible cohorts.

Common validation messages

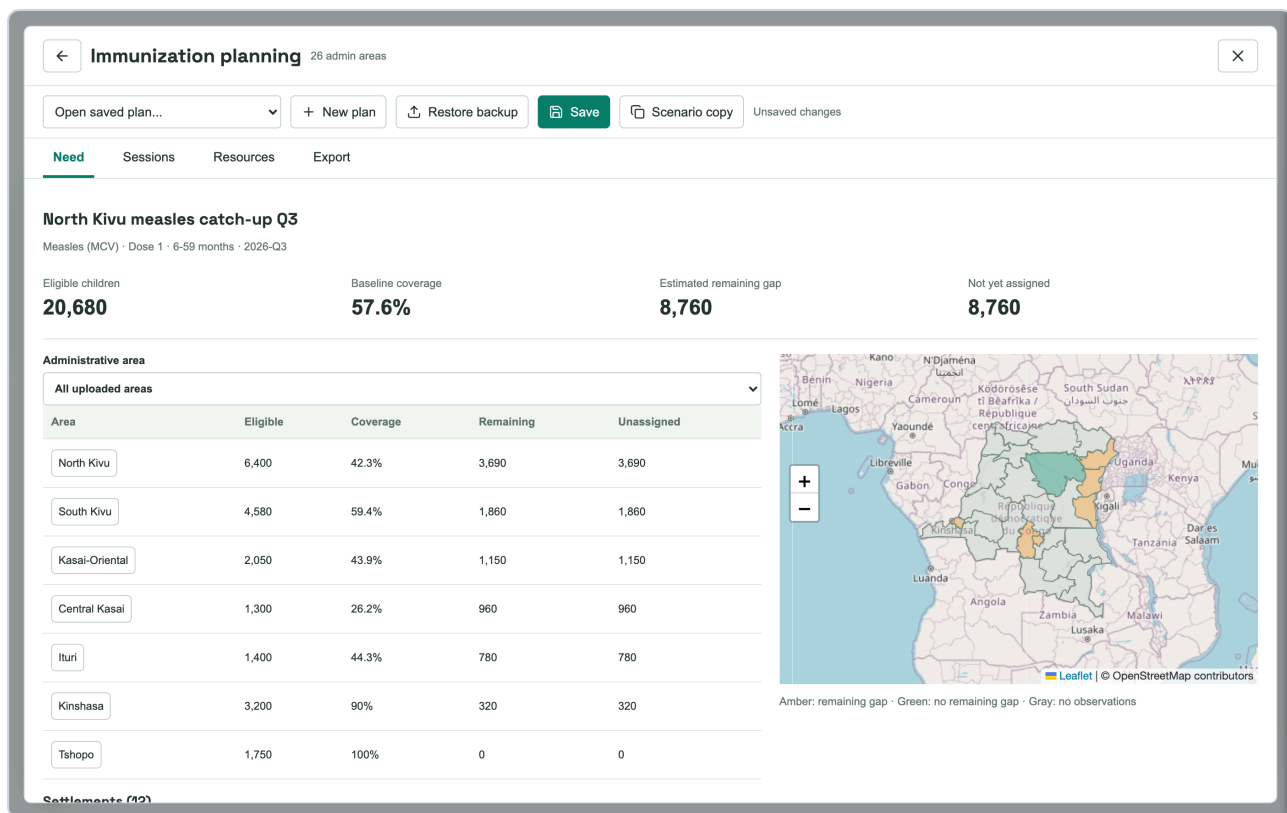
Validation message	What it means and how to fix it
No matching administrative area	The area code or coordinates did not fall inside any loaded boundary. Check the join field, the code values, or whether your boundary file covers that location.
Ambiguous area; use unique area codes	The code matched more than one boundary. Join on a field that is unique per area.
Coordinates conflict with area code	The point lies outside the polygon of the coded area. One of the two is wrong — resolve before importing.
Invalid coordinates	Latitude/longitude present but not usable (text, or out of range). Use decimal degrees.
Target / vaccinated must be a non-negative whole number	Counts must be whole numbers ≥ 0 . A zero is valid data; a blank is not.
Vaccinated exceeds target; review denominator	More children vaccinated than eligible usually means an outdated denominator or catchment overlap. Fix the source data.
Duplicate settlement observation	The same settlement ID appears twice in one area. Merge or re-code the rows.

1. Review the excluded rows (if any). Excluded rows never contribute to totals.
2. Tick the confirmation checkbox, then click **Create plan from N valid rows**.

Zeroes are preserved

A settlement with zero vaccinated children imports as exactly that — a real observation of zero, distinct from missing data. Do not delete zero rows; they often mark the settlements that most need sessions.

7 Review need: coverage and gaps



The Need tab: headline totals, per-area rollup, and the map. Amber areas have a remaining gap, green areas have none, gray areas have no observations.

- **Baseline coverage** is computed from summed counts — total vaccinated divided by total eligible — never by averaging percentages. A large settlement at 10% and a small one at 90% roll up to what the counts actually say.
- **Estimated remaining gap** = eligible – vaccinated – delivered so far (from sessions).
- **Not yet assigned** = the gap minus children already assigned to planned sessions. This is your to-do number.

Click an area in the table or on the map to filter the settlement list to that area. The settlement table is sorted by remaining gap and paginated at 100 rows, so the biggest gaps are always on the first page. Use each row's **Plan session** button to jump straight to the Sessions tab with that settlement pre-selected.

Need is not feasibility

The Need tab shows where unvaccinated children are — it says nothing about access, security, or cold chain. Keep high-need but hard-to-reach settlements visible in the plan rather than dropping them; record constraints in the session notes and review them operationally.

8 Plan sessions and record delivery

1. Pick a settlement (the dropdown shows each one's unassigned count), a date, a team name, and a delivery mode — **Fixed**, **Outreach**, or **Mobile**.
2. Enter the number of children assigned and click **Add session**. The app blocks allocations beyond a settlement's baseline gap, so you cannot plan for children who do not need the dose.
3. After sessions happen, enter the **Delivered** count on each row. Delivered children immediately reduce the remaining gap on the Need tab — that is the reconciliation loop.

The screenshot shows the 'Immunization planning' interface. At the top, there's a header with a back arrow, the title 'Immunization planning', and '26 admin areas'. Below this is a toolbar with buttons: 'Open saved plan...', '+ New plan', 'Restore backup', 'Save', 'Scenario copy', and 'Unsaved changes'. A tab bar below the toolbar has 'Need', 'Sessions' (active), 'Resources', and 'Export'. The main content area is titled 'North Kivu measles catch-up Q3' with a subtitle 'Measles (MCV) · Dose 1 · 6-59 months · 2026-Q3'. Below this are form fields for 'Settlement' (dropdown showing 'Kasai-Oriental / Mbuji-Mayi - Diulu (750 unassigned)'), 'Date' (text input '09/22/2026'), 'Team' (text input 'Equipe fixe Kasai 1'), 'Delivery mode' (dropdown showing 'Fixed'), and 'Children assigned' (text input). A green '+ Add session' button is below these fields. A note states: 'Each session allocation must cover different children within the settlement. Delivered counts must be unique children receiving this planned dose.' Below the note is a table with columns: Settlement, Date, Team, Mode, Planned, Delivered, and an action icon. The table has two rows: 1. Settlement: Goma - Majengo, Date: 2026-09-21, Team: Equipe mobile Nord-Kivu 1, Mode: Outreach, Planned: 300, Delivered: 260 (highlighted with a green box), Action: trash icon. 2. Settlement: Mbuji-Mayi - Diulu, Date: 2026-09-22, Team: Equipe fixe Kasai 1, Mode: Fixed, Planned: 400, Delivered: 0, Action: trash icon.

Settlement	Date	Team	Mode	Planned	Delivered	
Goma - Majengo	2026-09-21	Equipe mobile Nord-Kivu 1	Outreach	300	260	
Mbuji-Mayi - Diulu	2026-09-22	Equipe fixe Kasai 1	Fixed	400	0	

Two sessions planned for different teams and dates; the first already has 260 of 300 children recorded as delivered.

Count unique children, not doses given

Each session allocation must cover different children within the settlement, and delivered counts must be unique children receiving this plan's dose. Double-counting here silently inflates coverage downstream.

9 Estimate doses, teams, and budget

Enter your program's assumptions; results update instantly. All formulas are shown in the app footer:

- **Doses** = $\lceil \text{planned children} \div (1 - \text{wastage}) \times (1 + \text{buffer}) \rceil$, one planned dose per child.
- **Additional doses needed** = doses – available stock (never below zero).
- **Team-days** = for each team and date, $\lceil \text{assigned children} \div \text{children per team-day} \rceil$, summed.
- **Budget** = team-days × cost per team-day, in your chosen currency.

The screenshot shows the 'Immunization planning' app interface. At the top, there's a header with a back arrow, the title 'Immunization planning', and '26 admin areas'. Below the header is a toolbar with buttons: 'Open saved plan...', '+ New plan', 'Restore backup', 'Save', 'Scenario copy', and 'Unsaved changes'. A tab bar at the bottom of the toolbar shows 'Need', 'Sessions', 'Resources' (selected), and 'Export'. The main content area is titled 'North Kivu measles catch-up Q3' with a subtitle 'Measles (MCV) - Dose 1 - 6-59 months - 2026-Q3'. Under 'Scenario assumptions', there are input fields for 'Children per team-day' (150), 'Wastage (%)' (15), 'Supply buffer (%)' (10), 'Available doses' (200), 'Cost per team-day' (120), and 'Budget currency' (USD). Below these, a summary table shows: 'Doses including wastage/buffer' (906), 'Additional doses needed' (706), 'Required team-days' (5), and 'Team-day budget (USD)' (600). An amber banner at the bottom states: '2 team/date assignments exceed daily capacity. Add teams or reschedule before operational review.' Below the banner, a small text block explains the formulas used.

Children per team-day	Wastage (%)	Supply buffer (%)
150	15	10

Available doses	Cost per team-day	Budget currency
200	120	USD

Doses including wastage/buffer	Additional doses needed	Required team-days	Team-day budget (USD)
906	706	5	600

2 team/date assignments exceed daily capacity. Add teams or reschedule before operational review.

Doses = ceiling(planned children / (1 - wastage) × (1 + buffer)). One planned dose per child. Team-days are rounded up for each team/date workload. Budget covers team-days only; transport, cold-chain equipment, vial presentation, supplies, and access still require operational review.

The amber banner flags team/date workloads above daily capacity — here, both sessions exceed the 150-children capacity and need extra teams or extra days.

What the budget does not include

The budget covers team-days only. Transport, cold-chain equipment, vial presentation, supplies, and access costs still require operational review — the export states this on every workbook.

Compare scenarios

Use **Scenario copy** in the toolbar to duplicate the plan, change assumptions (capacity, wastage, allocations), and compare the resource totals side by side. Scenario copies are independent plans and save separately.

10 Save, export, and back up

1. **Save** stores the plan in this browser on this device, with a version number that increases on each save. The toolbar always shows whether you have unsaved changes.
2. On the **Export** tab, generate the **Planning workbook** — an Excel file with the plan metadata and assumptions, district rollups, settlement gaps, session allocations, actual delivery, and resource totals.
3. **Print session plan** produces a one-page schedule for field teams (allow pop-ups if the browser blocks it).
4. **Backup JSON** downloads the full plan as a file. Restore it later with **Restore backup** — against the same boundary set — for transfer to another machine or safekeeping.

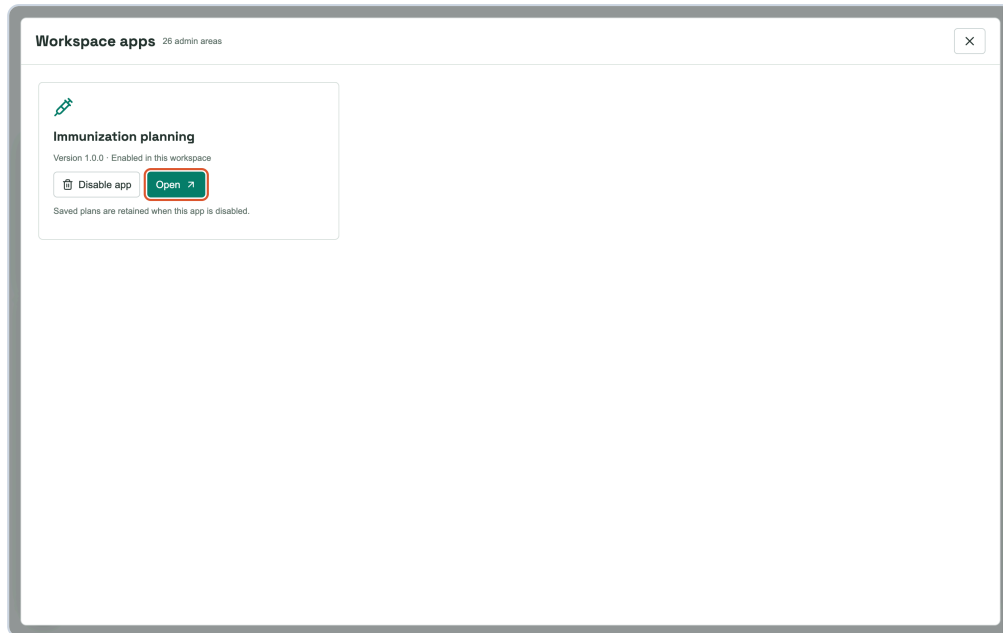
The screenshot shows the 'Immunization planning' interface. At the top, there's a title bar with a back arrow, 'Immunization planning', and '26 admin areas'. Below this is a toolbar with buttons: 'Open saved plan...', '+ New plan', 'Restore backup', 'Save', and 'Scenario copy'. A status indicator says 'Saved on this device · v1'. Below the toolbar are tabs: 'Need', 'Sessions', 'Resources', and 'Export' (which is active). The main content area is titled 'North Kivu measles catch-up Q3' with a subtitle 'Measles (MCV) · Dose 1 · 6-59 months · 2026-Q3'. It contains a form with three columns: 'Plan name' (North Kivu measles catch-up Q3), 'Vaccine / antigen' (Measles (MCV)), and 'Dose' (1). Below this are 'Eligible age / cohort' (6-59 months), 'Observation period' (2026-Q3), and 'Data source / revision' (EPI district reports, Aug 2026). At the bottom of the form are three buttons: 'Planning workbook' (highlighted with a red box), 'Print session plan', and 'Backup JSON'. A small note at the very bottom states: 'Workbook includes district totals, settlement gaps, session allocations, actual delivery, resources, source, assumptions, and revision. Plans are saved on this browser/device. A backup can be restored against the same uploaded boundaries.'

Exports carry the source, assumptions, revision number, and import audit (rows accepted and excluded), so reviewers can trace every figure.

Browser storage is not a server

Plans live in your browser's local storage. Clearing site data deletes them, and colleagues cannot see them. For anything you would mind losing, download the JSON backup and file the exported workbook.

Reopening a saved plan



Open the app again from Workspace apps, then pick the plan from **Open saved plan...** in the toolbar. Versions are shown next to each name.

11 Troubleshooting

Symptom	Cause and fix
Every imported row is rejected with several errors at once	Column misalignment — usually two header rows in the file or data pasted into the template in a different column order. Keep exactly one header row and import the file directly.
"No administrative boundaries loaded" in the app catalog	Upload your boundary file in the map workspace first, then reopen Workspace apps.
"This plan belongs to another boundary workspace"	The plan was created with a different boundary file. Reload that boundary file, or restore a JSON backup that matches the current one.
"This plan changed in another tab"	The same plan was saved from a second tab or window. Reopen the plan to get the latest version, or save your work as a scenario copy.
Save button stays disabled	There is nothing to save (no changes), or a validation error is shown at the top of the app — fix it first.
Print produces nothing	The browser blocked the pop-up window. Allow pop-ups for this site and try again.
"Browser storage unavailable"	Private-browsing mode or blocked site data. Use a normal window and allow site data for the application.

12 Data handling and limitations

- **Your data stays on your device.** Observations, sessions, and plans are processed and stored locally in the browser. Nothing in this app sends your settlement data to a server or to AI services.
- **Aggregate estimates, not registries.** The app identifies areas and settlements for action from aggregate counts. It does not — and cannot — identify individual eligible children. Do not import patient-level records.
- **Planning support, not certification.** The workflow follows the shape of WHO Reaching Every District (RED) microplanning practice: review local data, map need, plan sessions, estimate supplies, reconcile delivery. That is a design reference, not a WHO endorsement, and every plan still needs review by your program team.
- **Travel and access are not modeled.** Session dates and team workloads do not account for travel time or road access. Use the platform's logistics tools and local knowledge when scheduling.